

# **SNIP Building Codes and Standards**

## **LOADS AND EFFECTS**

### **SNiP 2.01.07-85**

Applying to design of building structures and foundations, the present norms define basic rules and regulations of dead and live loads and effects as well as their combinations.

Loads and effects on non-standard building structures and foundations are allowed to be calculated under some special technical conditions.

Note 1: 1. Further on, wherever it is possible, the term “effect” is omitted and substituted with the word “load” as well as “buildings and structures” are substituted with “structures”.

2. In the case of reconstruction design loads to be calculated according to examination results of the existing structures. At the same time, atmospheric loads may be adopted in accordance with the data provided by the State Hydrometeorological Committee.

#### **1. General**

1.1. The design should take into account the loads arising in the course of construction and operation of buildings as well as in the course of production, storage and transportation of construction materials.

1.2. The loads listed in the present norms are characterized with their standard values.

The load of a particular kind is usually characterized with a single standard value. For loads affecting floors of residential, public and farm buildings due to people, animals and equipment as well as bridge crane loads, overhead crane loads, snow loads and temperature effects it is set two standard values: full and low (the latter is taken into account if it is required to check load duration, endurance and in other cases stipulated in the design norms of building structures and foundations).

1.3. Design load should be calculated as the product of its standard value and the  $\gamma_f$  load safety factor. The safety factor conformed to the given limiting state ought to be taken as follows:

- a) in the case of analysis of strength and stability – according to items 2.2, 3.4, 3.7, 3.11, 4.8, 5.7, 6.11, 7.3 and 8.7;
- b) in the case of endurance analysis – equal to one;
- c) in the case of stiffness analysis – equal to one unless design norms of building structures and foundations imply other values;
- d) in the case of analysis of other limiting states – in accordance with design norms of building structures and foundations.

If statistics are available, design loads may be calculated directly by their specified exceedance probability.

Upon calculations of structures and foundations that define construction conditions of buildings and structures, the calculated values of snow, wind, sleet loads and temperature effects should be reduced by 20%.

If it is required to make analysis of strength and stability in the case of fire, explosion effects, transport collisions with structural parts, load safety factor should be taken equal to one for all the loads allowed.

Note: As regards the loads having two standard values, the corresponding design values are to be calculated by one and the same load safety factor (for the given limiting state).

## Classification of loads

- 1.4. Depending on load duration, loads may be divided into dead and live (sustained, short-term, special) loads.
- 1.5. The loads arising in the course of production, storage and transportation of structures as well as in the course of structure mounting should be adopted in calculations as short-term loads.  
The loads arising in the course of building operation should be taken in accordance with items 1.6-1.9.
- 1.6. Dead loads include the following:
  - a) weight of structural parts including weight of bearing and enclosing structures;
  - b) weight and pressure of ground (mounds and fillings), rock pressure.The prestressing forces remained in building structures and foundations should be adopted in calculations as dead load forces.
- 1.7. Sustained loads include the following:
  - a) weight of temporary partitions, grouts and concreting under equipment;
  - b) weight of fixed installations: machines, devices, engines, tanks, pipelines supplied with reinforcement, supports and insulation, belt conveyors, permanent hoists with their cables and guides as well as weight of liquids and solid substances filling the equipment;
  - c) pressure of gases, liquids and granular materials in tanks and pipelines; overpressure and air depression emerging in the course of shaft ventilation;
  - d) floor loads arising from stored materials and shelves in storerooms, cold stores, granaries, book depositories, archives and other similar rooms;
  - e) technological temperature effects of fixed installations;
  - f) weight of water layer of water-filled flat surfaces;
  - g) weight of occupational dust layer unless its accumulation is prevented with the relative measures;
  - h) loads due to people, animals and equipment affecting floors of residential, public and farm buildings having low standard values listed in Table No. 3;
  - i) vertical loads of bridge cranes and overhead cranes having a low standard value which is calculated by multiplication of the full standard value of the vertical load due to a single crane (see item 4.2) in each building span by the following coefficient: 0,5 – for groups of 4K-6K crane operation; 0,6 - for groups of 7K crane operation; 0,7 - for groups of 8K crane operation. The groups of crane operation to be adopted according to GOST 25546-82;
  - j) snow loads having a low standard value which is calculated by multiplication of the full standard value in accordance with the guidelines listed in item 5.1 by the following coefficient: 0,5 – for IV area; 0,6 – for V and VI areas;
  - k) temperature effects having low standard values that should be calculated in compliance with the guidelines listed in items 8.2-8.6 on the assumption of  $\theta_1 = \theta_2 = \theta_3 = \theta_4 = \theta_5 = 0$ ,  $\Delta_I = \Delta_{VII} = 0$ ;
  - l) effects caused by defrosting of frozen ground and by ground deformations unaccompanied by a critical change of the ground structure;
  - m) effects caused by humidity change, shrinkage and creep of materials.
- 1.8. Short-term loads include the following:
  - a) equipment loads arising in startup, transient and test modes as well as during equipment replacement or substitution;
  - b) weight of people and repair materials in equipment service and maintenance areas;
  - c) loads due to people, animals and equipment affecting floors of residential, public and farm buildings having full standard values excluding the loads listed in item 1.7,a,b,d,e;
  - d) loads caused by movable lifting and transport machinery (loaders, electric cars, piler cranes, telfers as well as bridge cranes and overhead cranes with a full standard value);
  - e) snow loads with a full standard value;
  - f) temperature effects with a full standard value;
  - g) wind loads;
  - h) sleet loads.

1.9. Special loads include the following:

- a) seismic effects;
- b) explosion effects;
- c) loads caused by abrupt breakdown in process, temporary malfunction or breakage of the equipment;
- d) effects caused by foundation deformations accompanied by a critical change of the ground structure (upon soaking of subsiding ground) or ground settlement in mining areas or karstic fields.

## **Load combinations**

1.10. Calculations of structures and foundations as regards limiting states of the first and second groups should be made considering adverse load combinations and their corresponding forces.

Such combinations are set according to real variant analysis of simultaneous effect of different loads for the given working stage of the structure or foundation.

1.11. Subject to the load structure allowed, there are the following load combinations:

- a) basic load combinations consisting of dead, sustained and short-term loads;
  - b) special load combinations consisting of dead, sustained, short-term and one of the special loads.
- Live loads with two standard values should be included into combinations as sustained loads – considering a low standard value, or as short-term loads – considering a full standard value.
- The special load combinations comprising explosion effects or loads caused by transport collisions with structural parts may exclude short-term loads stipulated in item 1.8.

1.12. Considering the combinations containing dead loads and not less than two live loads, calculated values of live loads or the relative forces should be multiplied by combination coefficients equal to the following figures:

In basic combinations for sustained loads  $\psi_1=0,95$ ; for short-term loads -  $\psi_2=0,9$ ;

In special combinations for sustained loads  $\psi_1=0,95$ ; for short-term loads -  $\psi_2=0,8$ , except for the cases stipulated in the design norms for seismic zones and in other design norms of structures and foundations. At the same time, special load is to be taken without reduction.

Considering the basic combinations containing dead loads and a live load (sustained or short-term),  $\psi_1$  and  $\psi_2$  coefficients should not be taken into account.

Note: In the basic combinations considering three or more short-term loads, their design values may be multiplied by  $\psi_2$  combination coefficient which is to be adopted for the first (as regards degree of effect) short-term load – 1,0, for the second – 0,8 and for the rest – 0,6.

1.13. Allowing for the load combinations in accordance with the guidelines of item 1.12, the following loads should be taken as a live load:

- a) load of a particular kind arising from a single source (pressure or depression in tank; snow, wind or sleet loads, temperature effects, loads caused by a loader, electric car, bridge crane or overhead crane);
- b) load arising from several sources if their concurrent effect is taken into account in the standard and design values (load due to equipment, people and stored materials transferred to one or several floors considering  $\psi_A$  and  $\psi_n$  coefficients specified in items 3.8 and 3.9; load caused by several bridge cranes or overhead cranes considering  $\psi$  coefficient stipulated in item 4.17; sleet and wind load calculated in accordance with item 7.4).

## **2. Weight of structures and grounds**

2.1. The standard weight of prefabricated structures should be calculated according to standards, shop drawings or ratings of manufacturing plants, other building structures and grounds - in compliance with design dimensions and specific gravity of materials and grounds considering their humidity degree in the course of construction and operation of structures.

2.2.  $\gamma_f$  load safety factors for the weight of building structures are listed in the Table 1 hereinbelow.

Table 1

| Building structures and types of ground                                                                                                                                                                                                                                                   | $\gamma_f$ load safety factor |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| <i>Structures:</i>                                                                                                                                                                                                                                                                        |                               |
| steel                                                                                                                                                                                                                                                                                     | 1,05                          |
| concrete (with the average consistency exceeding 1600 kg/m <sup>3</sup> ), reinforced concrete, masonry, reinforced masonry, wooden                                                                                                                                                       | 1,1                           |
| concrete (with the average consistency 1600 kg/m <sup>3</sup> and less), insulation, smooth and finishing layers (plates, rolled materials, fillings, screeds etc.) that are                                                                                                              |                               |
| prefabricated                                                                                                                                                                                                                                                                             | 1,2                           |
| produced at site                                                                                                                                                                                                                                                                          | 1,3                           |
| <i>Grounds:</i>                                                                                                                                                                                                                                                                           |                               |
| of natural deposit                                                                                                                                                                                                                                                                        | 1,1                           |
| fill-up ground                                                                                                                                                                                                                                                                            | 1,15                          |
| Notes: 1. In the case of structure stability test and in the other cases when weight reduction of structures and grounds may deteriorate structure operation, the calculations should be made taking $\gamma_f$ load safety factor for the weight of structures or its part equal to 0,9. |                               |
| c) When calculating ground loads, the loads transferred to the ground due to stored materials, equipment and transport means should be taken into account.                                                                                                                                |                               |
| 3. For steel structures which dead weight forces exceed 50% of the total force - $\gamma_f = 1,1$ .                                                                                                                                                                                       |                               |

## 2. Loads due to equipment, people, animals, stored materials and products

- 3.1. The norms of the present paragraph apply to the loads transferred to building floors and ground floors due to people, animals, equipment, products, materials and temporary partitions.

Alternative transfers of the mentioned loads to building floors should be adopted in accordance with the particular conditions of construction and operation of the buildings. If at the design stage the data of such conditions are not sufficient, it is necessary to consider the following alternative loadings of different floors when calculating structures and foundations:

complete loading; adverse partial loading upon calculation of the structures and foundations tolerant to such a loading pattern; lack of live load.

At the same time, the total live load transferred to multi-storeyed building floors under the adverse partial loading should not exceed the load under the complete loading calculated with consideration of  $\psi_n$  combination coefficients which values are to be calculated by formulas (3) and (4).

### Point loads and loads transferred to railings

- 3.2. The bearing members of floors, coverings, staircases and balconies (loggias) are to be checked as regards vertical point load transferred to the member in an adverse position on a square platform with the sides not exceeding 10 cm (under the lack of other live loads). Unless on the basis of technological solutions the building assignment provides higher standard values of point loads, they are to be equal to the following:

- for floors and staircases – 1,5 kN (150 kgf);
- for attic ceilings, coverings, terraces and balconies – 1,0 kN (100 kgf);
- for surfaces that may be walkable only by ladders and foot-bridges – 0,5 kN (50 kgf).

The members that are applied to local loads caused by equipment or transport means which may arise during installation or structure operation, may avoid testing as regards the mentioned point load.

- 3.3. The standard horizontal loads transferred to railings of staircases and balconies should be taken equal to the following:

- for residential buildings, infant schools, holiday homes, sanatoriums, hospitals and other health establishments – 0,3 kN/m (30 kgf/m);
- for stands and sports halls – 1,5 kN/m (150 kgf/m);
- for other buildings and rooms upon the lack of special requirements – 0,8 kN/m (80 kgf/m).

Table 3

| Building and rooms                                                                                                                                                                                                                                                    | Standard load values $p$ , kPa (kgf/m <sup>2</sup> ) |                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-------------------------|
|                                                                                                                                                                                                                                                                       | complete                                             | reduced                 |
| 1. Apartments of residential buildings, bedrooms of infant schools and boarding schools, living premises of holiday homes and boarding houses as well as hostels and hotels; wards of hospitals and sanatoriums; terraces                                             | 1,5 (150)                                            | 0,3 (30)                |
| 2. Offices and service rooms of administration, engineers and researchers of organizations and establishments; classrooms of educational institutions; amenity rooms (lockers, shower rooms, washrooms, WCs) of industrial plants, public buildings and constructions | 2,0 (200)                                            | 0,7 (70)                |
| 3. Health facilities and laboratories; laboratories of educational and scientific institutions; computer rooms; kitchens of public buildings; technical floors; basement premises                                                                                     | not less than 2,0 (200)                              | not less than 1,0 (100) |
| 4. Halls:                                                                                                                                                                                                                                                             |                                                      |                         |
| a) reading halls                                                                                                                                                                                                                                                      | 2,0 (200)                                            | 0,7 (70)                |
| b) dining-rooms (in cafes, restaurants, canteens)                                                                                                                                                                                                                     | 3,0 (300)                                            | 1,0 (100)               |
| c) meeting-rooms, waiting-rooms, auditoriums and concert halls; sports halls                                                                                                                                                                                          | 4,0 (400)                                            | 1,4 (140)               |
| d) trading floors, show-rooms and expositions                                                                                                                                                                                                                         | not less than 4,0 (400)                              | not less than 1,4 (140) |
| 5. Book depositories and archives                                                                                                                                                                                                                                     | not less than 5,0 (500)                              | not less than 5,0 (500) |
| 6. Show stages                                                                                                                                                                                                                                                        | not less than 5,0 (500)                              | not less than 1,8 (180) |
| 7. Stands:                                                                                                                                                                                                                                                            |                                                      |                         |
| a) with fixed seats                                                                                                                                                                                                                                                   | 4,0 (400)                                            | 1,4 (140)               |
| b) for standing audience                                                                                                                                                                                                                                              | 5,0 (500)                                            | 1,8 (180)               |
| 8. Attic premises                                                                                                                                                                                                                                                     | 0,7 (70)                                             | -                       |
| 9. Surface cover in the following areas:                                                                                                                                                                                                                              |                                                      |                         |
| a) in potentially crowded areas (exits from production areas, halls and rooms etc.)                                                                                                                                                                                   | 4,0 (400)                                            | 1,4 (140)               |
| b) in recreation areas                                                                                                                                                                                                                                                | 1,5 (150)                                            | 0,5 (50)                |
| c) other areas                                                                                                                                                                                                                                                        | 0,5 (50)                                             | -                       |
| 10. Balconies (loggias) considering the following load:                                                                                                                                                                                                               |                                                      |                         |
| a) strip uniform load in the area 0,8 m wide along balcony (loggia) railing                                                                                                                                                                                           | 4,0 (400)                                            | 1,4 (140)               |
| b) continuous uniform load on balcony (loggia) which effect is more adverse than the one calculated under the previous item 10,a                                                                                                                                      | 2,0 (200)                                            | 0,7 (70)                |
| 11. Equipment maintenance areas in production premises                                                                                                                                                                                                                | not less than 1,5 (150)                              | -                       |
| 12. Entrance halls, foyers, corridors and staircases (including relevant passageways) that adjoin the rooms listed in the following items:                                                                                                                            |                                                      |                         |
| a) 1, 2 and 3                                                                                                                                                                                                                                                         | 3,0 (300)                                            | 1,0 (100)               |
| b) 4, 5, 6 and 11                                                                                                                                                                                                                                                     | 4,0 (400)                                            | 1,4 (140)               |
| c) 7                                                                                                                                                                                                                                                                  | 5,0 (500)                                            | 1,8 (180)               |
| 13. Station platforms<br>small cattle                                                                                                                                                                                                                                 | 4,0 (400)                                            | 1,4 (140)               |

Notes: 1. The loads specified in item 8 should be taken into account as regards the area free from equipment and materials.  
 2. The loads stipulated in item 9 are to be considered without snow load.  
 3. The loads specified in item 10 ought to be considered upon calculation of balcony (loggia) bearing structures and their restraining wall areas. Upon calculation of underlying wall area, foundations and basements, balcony (loggia) loads should be adopted as equal to the loads of basic adjoining rooms. The loads in view are to be consequently reduced considering the guidelines stated in items 3.8 and 3.9.  
 Standard load values for the buildings and rooms listed in items 3, 4.d, 5, 6, 11 and 14 should be taken under building assignment and in accordance with technological solutions.

#### 4. Snow loads

- 4.1. The complete standard value of snow load transferred to the horizontal projection of surface  $s$  is to be calculated by the following formula:

$$s = s_0 \mu \quad (5)$$

where  $s_0$  – standard snow cover weight on 1 m<sup>2</sup> of horizontal ground surface; the mentioned value is to be adopted according to item 5.2;

$\mu$  – factor of conversion from snow cover weight to snow load transferred to the roof that is adopted according to items 5.3-5.6.

- 4.2. Standard value of snow cover weight  $s_0$  on 1 m<sup>2</sup> of horizontal ground surface should be adopted subject to the snow region of USSR according to Table 4.  
 4.3. Snow load distribution patterns and  $\mu$  factor values are to be adopted in compliance with obligatory Appendix No. 3, the intermediate values of  $\mu$  factor being set by simple interpolation. It is required to consider patterns of snow load effecting a half or a quarter of a span (for skylight roofs – in areas  $b$  wide) in case due to partial loading the structure members encounter more adverse state.

Note: If required, snow loads are to be calculated considering a further extension of the building.

- 4.4. The variants of high local snow loads listed in obligatory Appendix No. 3 should be considered upon calculations of slabs, coverings and roof spans as well as upon calculation of the bearing structure members (frames, beams, columns etc.) for that the specified variants define section dimensions.

Note: Upon calculation of structures it is possible to apply simplified patterns of snow loads equivalent to the load effects listed in obligatory Appendix No. 3. When calculating frames and columns of production buildings, it is allowed to consider only uniformly distributed snow load excluding roof level differences that require high snow load.

- 5.5\*. The coefficients  $\mu$  set according to schemes 1, 2, 5 and 6 of obligatory Appendix No. 3 - for flat roofs (with the slopes up to 12% or with  $\frac{f}{l} \leq 0,05$ ) of single-span and multispans skylightless buildings designed in the districts where the average wind speed for the three coldest months is  $v \geq 2$  m/sec – should be reduced by multiplication by the coefficient  $c_e = (1,2 - 0,1v\sqrt{k})(0,8 + 0,002b)$  where  $k$  is to be taken under Table 6;  $b$  – roof width taken as not exceeding 100 m.

For the roofs with the slopes from 12 up to 20% of single-span and multispans skylightless buildings designed in the districts where  $v \geq 4$  m/sec, the coefficient  $\mu$  set according to schemes 1 and 5 of obligatory Appendix No. 3 should be reduced by multiplication by the coefficient equal to 0,85.

The average wind speed  $v$  for the three coldest months is to be adopted under chart 5 of obligatory Appendix No. 5.

Snow load reduction allowed by the present paragraph does not cover the following issues:

- roofs of the buildings located in the regions where the average monthly temperature in January is above -5°C (see chart 5 of obligatory Appendix No. 5);
- roofs of the buildings protected from the direct wind with higher neighboring buildings located at a distance less than  $10h_1$  where  $h_1$  – height difference between the neighbouring and designed buildings;
- roof areas  $b$ ,  $b_1$  and  $b_2$  wide near height level differences between the buildings and parapets (see schemes 8-11 of obligatory Appendix No. 3).

Table 4

| Snow regions of USSR<br>(adopted according to<br>chart 1 of obligatory<br>Appendix No. 5)                                                                                                                                                                                                                                                                                                                                                                                                                                            | I        | II       | III       | IV        | V         | VI        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|-----------|-----------|-----------|-----------|
| $s_0, \text{kPa (kgf/m}^2\text{)}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0,5 (50) | 0,7 (70) | 1,0 (100) | 1,5 (150) | 2,0 (200) | 2,5 (250) |
| Note: In highlands and insufficiently explored regions specified in chart 1 of obligatory Appendix No. 5 as well as in points exceeding 1500 m above the sea level and in places of complex relief standard weight of snow cover should be set according to the data submitted by Goskomgidromet. At that, the average volume of yearly maximum water resources defined upon the snow survey being made for not less than 10 years at the spot protected from wind is to be adopted as the standard weight value of snow cover $s_0$ |          |          |           |           |           |           |

5.6. When calculating snow loads for unheated roofs of high heat-evolution workshops with their roof slopes exceeding 3%, coefficients  $\mu$  are to be reduced by 20% despite the reduction specified in item 5.5 to provide a proper drainage of water from melted snow.

5.7. As regards snow load,  $\gamma_f$  load safety factor is assumed as equal to 1,4. Upon calculation of roof members for which the ratio between the uniformly distributed standard load due to the roof weight (including fixed equipment weight) and the snow cover standard weight  $s_0$  is less than 0,8,  $\gamma_f$  is to be assumed as 1,6.

## 6. Wind loads

6.1. Wind load is considered as the *set* of the following:

- normal pressure  $w_e$  which is applied to the surface of structure or member;
- frictional force  $w_f$  directed at a tangent to the surface and relative to the area of its horizontal (regarding double ridge or corrugated roofs as well as skylight roof) or vertical projection (walls supplied with loggias or similar structures);
- normal pressure  $w_i$  which is applied to the internal surfaces supplied with permeable enclosures as well as opening or constantly opened apertures;

or as the *normal pressure*  $w_x, w_y$  resulting from the total strength of the construction along axes  $x$  and  $y$  and conditionally applied to the construction plan projection which is at right angle to the axis.

Upon calculation of high structures which relative dimensions meet the condition of  $h/d > 10$ , it is required to make a verifying calculation of vertical excitation (wind resonance);  $h$  – structure height,  $d$  – minimum cross section at level  $2/3h$ .

6.2. Wind load is to be calculated as the sum of average and pulsating components.

Upon calculation of internal pressure  $w_i$  as well as upon calculation of high buildings up to 40 m and one-storey buildings up to 36 m – in case the ratio between the height and the span is less than 1,5 – that are located in areas of A and B types (see item 6.5), the pulsating component of wind load may be omitted.

6.3. The standard average component of wind load  $w_m$  at the height of  $z$  over the ground surface is to be calculated under the following formula:

$$w_m = w_0 k c \quad (6)$$

where  $w_0$  – standard wind pressure (see item 6.4)

$k$  – coefficient of wind pressure change in height (see item 6.5)

$c$  – aerodynamic coefficient (see item 6.6).

6.4. The standard wind pressure  $w_0$  should be adopted in accordance with the relative wind region of USSR under Table 5.

As regards mountainous and insufficiently explored regions marked at map 3, the standard wind pressure  $w_0$  may be set according to the data provided by the weather stations of Goskomgidromet as well as building areas survey with assumption of field experience. At that, the standard wind pressure  $w_0$  is to be calculated under the following formula:

$$w_0 = 0,61 v_0^2 \quad (7)$$

where  $v_0$  – numerically equal to a wind speed (m/sec) at the height of 10 m over the ground surface as regards the region of A type. The mentioned wind speed corresponds to 10-minute averaging interval and is exceeded once in 5 years on average (unless the technical conditions approved in accordance with established procedure consider other intervals of wind speed repetition).

6.5. The coefficient  $k$  considering wind pressure change in height  $z$  is set under table 6 subject to area type.

The following area types are adopted:

A – open coasts of seas, lakes and storage ponds as well as deserts, steppes, forest-steppes and tundra;

B – urban territory, woodland and other lands covered with obstructions higher than 10 m;

C – urban territory holding buildings exceeding 25 m high.

Table 5

| Wind regions of USSR (to be adopted according to map 3 of the obligatory Appendix No. 5) | Ia        | I         | II        | III       | IV        | V         | VI        | VII       |
|------------------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| $w_0$ , kPa (kgf/m <sup>2</sup> )                                                        | 0,17 (17) | 0,23 (23) | 0,30 (30) | 0,38 (38) | 0,48 (48) | 0,60 (60) | 0,73 (73) | 0,85 (85) |

The building may be considered as located in the given area if that area stretches on the windward side of the building at the distance of  $30h$  – at the building height  $h$  up to 60 m and 2 km – at a bigger height.

Table 6

| Height $z$ (m) | Coefficient $k$ for area types |      |      |
|----------------|--------------------------------|------|------|
|                | A                              | B    | C    |
| $\leq 5$       | 0,75                           | 0,5  | 0,4  |
| 10             | 1,0                            | 0,65 | 0,4  |
| 20             | 1,25                           | 0,85 | 0,55 |
| 40             | 1,5                            | 1,1  | 0,8  |
| 60             | 1,7                            | 1,3  | 1,0  |
| 80             | 1,85                           | 1,45 | 1,15 |
| 100            | 2,0                            | 1,6  | 1,25 |
| 150            | 2,25                           | 1,9  | 1,55 |
| 200            | 2,45                           | 2,1  | 1,8  |
| 250            | 2,65                           | 2,3  | 2,0  |
| 300            | 2,75                           | 2,5  | 2,2  |
| 350            | 2,75                           | 2,75 | 2,35 |
| $\geq 480$     | 2,75                           | 2,75 | 2,75 |

Notes: Upon calculation of wind load the area types may vary for different design wind direction.

6.6. The calculation of wind load components  $w_e$ ,  $w_f$ ,  $w_i$ ,  $w_x$ ,  $w_y$  requires use of the corresponding aerodynamic coefficients – external pressure  $c_e$ , friction  $c_f$ , internal pressure  $c_i$  and point resistance  $c_x$  or  $c_y$  that are adopted under the obligatory Appendix No. 3 where wind direction is pointed. Plus of  $c_e$  and  $c_i$  corresponds to a direction of wind pressure onto the relative surface; minus corresponds to a direction of wind pressure from the surface. Intermediate loads are to be defined by the simple interpolation.

The calculation of enclosing members' connections to bearing structures in the building corners and on the external contour of the roof requires assuming a localized negative wind pressure with aerodynamic coefficient  $c_e = -2$ , the given wind pressure being distributed upon the surfaces at the width of 1,5 m (draft 1).

In the cases beyond the obligatory Appendix No. 4 (other construction forms; assumption of other wind stream directions, on a proper ground, or components of total body resistance as regards other



directions etc.) aerodynamic coefficients may be adopted according to reference or experimental data or on the ground of construction model air blowing in wind tunnels.

Note: It is required to apply aerodynamic coefficients of external pressure  $c_e$  or point resistance  $c_x$  when calculating wind load transferred to the surfaces of internal walls and partitions if the external enclosure lacks (at the installation stage).

#### Draft 1. Zones of increased negative wind pressure

6.7. The standard pulsating component of wind load  $w_p$  at the height  $z$  is to be set for the following:

- a) for constructions (and their structural members) which first frequency of natural oscillations  $f_1$  (Hz) exceeds the limiting value of natural frequency  $f_l$  (see item 6.8) – by the following formula

$$w_p = w_m \zeta v \quad (8)$$

where  $w_m$  – to be set according to item 6.3;

$\zeta$  – coefficient of wind pressure pulsations at the height  $z$  that is to be adopted under table 7;

$v$  – coefficient of spatial correlation of wind pressure pulsations (see item 6.9);

Table 7

| Height $z$ (m) | Wind pressure pulsation coefficient $\zeta$ for area types |      |      |
|----------------|------------------------------------------------------------|------|------|
|                | A                                                          | B    | C    |
| $\leq 5$       | 0,85                                                       | 1,22 | 1,78 |
| 10             | 0,76                                                       | 1,06 | 1,78 |
| 20             | 0,69                                                       | 0,92 | 1,50 |
| 40             | 0,62                                                       | 0,80 | 1,26 |
| 60             | 0,58                                                       | 0,74 | 1,14 |
| 80             | 0,56                                                       | 0,70 | 1,06 |
| 100            | 0,54                                                       | 0,67 | 1,00 |
| 150            | 0,51                                                       | 0,62 | 0,90 |
| 200            | 0,49                                                       | 0,58 | 0,84 |
| 250            | 0,47                                                       | 0,56 | 0,80 |
| 300            | 0,46                                                       | 0,54 | 0,76 |
| 350            | 0,46                                                       | 0,52 | 0,73 |
| $\geq 480$     | 0,46                                                       | 0,50 | 0,68 |

#### Draft 2. Dynamic coefficients

1 – for reinforced concrete and stone structures as well as for steel framed buildings if enclosing structures are available ( $\delta = 0,3$ ); 2 – for steel towers, masts, lined chimneys, column-type devices including ones on reinforced concrete bases ( $\delta = 0,15$ ).

- b) for constructions (and their structural members) that may be considered as a system of a single freedom (cross frames of one-storey production buildings, water towers etc.) if  $f_1 < f_l$  – by the following formula

$$w_p = w_m \xi \zeta v \quad (9)$$

where  $\xi$  – dynamic coefficient which is defined according to draft 2 subject to parameter

$$\varepsilon = \frac{\sqrt{\gamma_f w_0}}{940 f_1} \text{ and logarithmic decrement of oscillation } \delta \text{ (see item 6.8);}$$

$\gamma_f$  – load safety factor (see item 6.11);

$w_0$  – standard wind load, Pa (see item 6.4);

- c) for buildings that are symmetric on plan and for that  $f_1 < f_l$ ; besides, for all constructions where  $f_1 < f_l < f_2$  (where  $f_2$  is the second frequency of natural oscillation) – by the following formula

$$w_p = m \xi \psi v \quad (10)$$

where  $m$  – construction weight at the height  $z$  that is related to the surface area affected by wind load;

$\xi$  – dynamic coefficient (see item 6.7,b);

$y$  – horizontal displacement of construction at the height  $z$  by the first type of natural oscillation (for symmetric-on-plan buildings of a constant height the displacement due to uniformly distributed horizontally imposed static load may be adopted as  $y$ );

$\psi$  – coefficient which is calculated by division of the construction into  $r$  zones; within the given zones wind load is to be adopted as constant by the following formula

$$\psi = \frac{\sum_{k=1}^r y_k w_{pk}}{\sum_{k=1}^r y_k^2 M_k} \quad (11)$$

where  $M_k$  – weight of  $k$  construction part;

$y_k$  – horizontal displacement of  $k$  centre;

$w_{pk}$  – resultant force of the wind load pulsating component for  $k$  construction part; that is calculated by formula (8).

As regards multi-storeyed buildings of constant height, rigidity and weight as well as a constant width of their windward side, the standard pulsating component of wind load at a height of  $z$  may be calculated by the following formula

$$W_p = 1.4 \frac{z}{h} \xi W_{ph} \quad (12)$$

where  $W_{ph}$  – the standard value of pulsating component of the wind load at the height  $h$  of the top calculated by formula (8).

**6.8.** It is necessary to determine limit value of the own vibration frequency  $f_l$ , hertz, by which it is possible to consider the inertia force that appear during vibration in compliance with corresponding proper form according to table 8.

Table 8

| Wind regions of USSR<br>(taken according to the map 3<br>of obligatory Annex 5) | $f_l$ , hertz, if |                 |
|---------------------------------------------------------------------------------|-------------------|-----------------|
|                                                                                 | $\delta = 0.3$    | $\delta = 0.15$ |
| Ia                                                                              | 0.85              | 2.6             |
| II                                                                              | 0.95              | 2.9             |
| III                                                                             | 1.1               | 3.4             |
| IV                                                                              | 1.2               | 3.8             |
| V                                                                               | 1.4               | 4.3             |
| VI                                                                              | 1.6               | 5.0             |
| VII                                                                             | 1.7               | 5.6             |
|                                                                                 | 1.9               | 5.9             |

The value of logarithmic decrement of vibrations  $\delta$  is to be taken

- for reinforced concrete and stone structures as well as for buildings with steel framework if there are walling structures  $\delta = 0.3$ ;
- for steel towers, masts, lined chimneys, column means including the ones on the reinforced concrete pedestals  $\delta = 0.15$ .

**6.9.** The coefficient of spatial correlation of pressure pulsation  $\nu$  must be calculated for design surface of the structure where the correlation of the pulsation is taken into account.

The design surface involves those parts of the surfaces of windward, leeward, lateral walls, roofs and structures like that from which the wind load is given to the calculated member of the structure.

If the design surface is close to be a rectangle which is oriented so that its sides are parallel to the main axes (draft 3), so the coefficient  $v$  must be determined according to the table 9 depending on the parameters  $\rho$  and  $\chi$  taken according to the table 10.

**Draft 3. Coordinate system when calculating the correlation coefficient  $v$ .**

$\chi$  is the wind direction.

Table 9

| $\rho$ , m | The coefficient $v$ if $\chi$ is in meters |      |      |      |      |      |      |
|------------|--------------------------------------------|------|------|------|------|------|------|
|            | 5                                          | 10   | 20   | 40   | 80   | 160  | 350  |
| 0.1        | 0.95                                       | 0.92 | 0.88 | 0.83 | 0.76 | 0.67 | 0.56 |
| 5          | 0.89                                       | 0.87 | 0.84 | 0.80 | 0.73 | 0.65 | 0.54 |
| 10         | 0.85                                       | 0.85 | 0.81 | 0.77 | 0.71 | 0.64 | 0.53 |
| 20         | 0.80                                       | 0.80 | 0.76 | 0.73 | 0.68 | 0.61 | 0.51 |
| 40         | 0.72                                       | 0.72 | 0.70 | 0.67 | 0.63 | 0.57 | 0.48 |
| 80         | 0.63                                       | 0.63 | 0.61 | 0.59 | 0.56 | 0.51 | 0.44 |
| 160        | 0.53                                       | 0.53 | 0.52 | 0.50 | 0.47 | 0.44 | 0.38 |

Table 10

| Main coordinate space parallel to which the design surface is | $\rho$ | $\chi$ |
|---------------------------------------------------------------|--------|--------|
| $zoy$                                                         | $b$    | $h$    |
| $zox$                                                         | $0.4a$ | $h$    |
| $xoy$                                                         | $b$    | $a$    |

When calculating the whole structure the dimensions of the design surface must be determined considering the obligatory Annex 4 instructions, at the same time for latticed structures it is necessary to take the measurements of design surface according to the outside bounding of the structure.

**6.10.** For the structures where  $f_2 < f_1$  it is necessary to make dynamic calculations considering  $s$  of the first forms of natural vibrations. The value  $s$  is to be calculated by the condition

$$f_s < f_l < f_{s+1}$$

**6.11.** Safety factor regarding the wind load  $\gamma_f$  is to be taken as equal to 1.4.

## 8. Temperature Climatic Effects

**8.1.** In cases provided in design norms documents it is necessary to consider the variety of average temperature at the time of  $\Delta t$  and temperature difference  $\vartheta$  in the section of the member.

**8.2.** Standard values of the variety of average temperature in the section of member during warm  $\Delta t_w$  and cold  $\Delta t_c$  season must be calculated by the following formulas:

$$\Delta t_w = t_w - t_{0c} \quad (15)$$

$$\Delta t_c = t_s - t_{0w} \quad (16)$$

where  $\Delta t_w$ ,  $\Delta t_c$  – standard values of average temperatures on the section of member during warm and cold season taken in compliance with item 8.3;

$t_{0w}$ ,  $t_{0c}$  – initial temperature during warm and cold season taken in compliance with item 8.6.

**8.3.** Standard values of average temperatures  $t_w$  and  $t_s$  and temperature difference in the section of member during warm  $\mathcal{G}_w$  and cold  $\mathcal{G}_c$  seasons for one-layer structures must be taken according to table 15.

Note. For multi-layer structures  $t_w$ ,  $t_s$ ,  $\mathcal{G}_w$ ,  $\mathcal{G}_c$  are calculated. The structures made of several kinds of material which have resembling thermal characteristics can be considered as one-layer ones.

**8.4.** Mean daily temperatures of outside air during warm  $t_{ew}$  and cold  $t_{ec}$  seasons must be calculated by the following formulas:

$$t_{ew} = t_{VII} + \Delta_{VII} \quad (17)$$

$$t_{ec} = t_I - \Delta_I \quad (18)$$

where  $t_I$ ,  $t_7$  – long-term monthly mean temperatures in January and July, taken in compliance with maps 5 and 6 of the obligatory Annex 5;

$\Delta_I, \Delta_7$  – deviation of mean daily temperature from the monthly mean temperatures ( $\Delta_I$  is taken according to table 7 of the obligatory Annex 5,  $\Delta_{VII} = 6$  degrees Celsius).

Table 15

| Structures of the building                                                                        | Buildings and structures in use                                               |                                                      |                                                                                 |
|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------------------|
|                                                                                                   | Non-heated buildings (without technological heat sources) and open structures | Heated buildings                                     | The buildings with artificial climate or with steady technological heat sources |
| The structures which are not defended from solar radiation (including outside walling structures) | $t_w = t_{ew} + \theta_1 + \theta_4$                                          |                                                      | $t_w = t_{iw} + 0.6(t_{ew} - t_{iw}) + \theta_2 + \theta_4$                     |
|                                                                                                   | $\mathcal{G}_w = \theta_5$                                                    |                                                      | $\mathcal{G}_w = 0.8(t_{ew} - t_{iw}) + \theta_3 + \theta_5$                    |
|                                                                                                   | $t_c = t_{ec} - 0.5\theta_1$                                                  | $t_c = t_{ic} + 0.6(t_{ec} - t_{ic}) - 0.5\theta_2$  |                                                                                 |
|                                                                                                   | $\mathcal{G}_c = 0$                                                           | $\mathcal{G}_c = 0.8(t_{ec} - t_{ic}) - 0.5\theta_3$ |                                                                                 |
| The structures which are defended from solar radiation (including inside structures)              | $t_w = t_{ew}$                                                                |                                                      | $t_w = t_{iw}$                                                                  |
|                                                                                                   | $\mathcal{G}_w = 0$                                                           |                                                      |                                                                                 |
|                                                                                                   | $t_c = t_{ec}$                                                                | $t_c = t_{ic}$                                       |                                                                                 |
|                                                                                                   | $\mathcal{G}_c = 0$                                                           |                                                      |                                                                                 |

*Symbols taken in table 15:*

$t_{ew}$ ,  $t_{ec}$  – mean daily temperature of outside air correspondingly during warm and cold seasons, taken in compliance with item 8.4;

$t_{iw}$ ,  $t_{ic}$  – temperatures of the air inside the rooms correspondingly during warm and cold seasons, taken according to GOST 12.1.005-88 or according to the building statement on basis of the technological solutions;

$\theta_1$ ,  $\theta_2$ ,  $\theta_3$  – increment of average in the section of member temperature and temperature difference from daily temperature variation of outside air, taken according to table 16;

$\theta_4$ ,  $\theta_5$  – increment of average in the section of member temperature and temperature difference from solar radiation, taken in compliance with item 8.5.

Notes: 1. If there is initial data about the temperature of the structure in stage of use of the building with steady technological heat sources, the values  $t_w$ ,  $t_c$ ,  $\mathcal{G}_w$ ,  $\mathcal{G}_c$  must be taken in compliance with this data.

2. For buildings and structure in stage of building  $t_w$ ,  $t_c$ ,  $\mathcal{G}_w$ ,  $\mathcal{G}_c$  are calculated as for the non-heated buildings in stage of use.

Table 16

| Structure of the building                                            | Temperature increment $\theta$ , degrees Celsius |            |            |
|----------------------------------------------------------------------|--------------------------------------------------|------------|------------|
|                                                                      | $\theta_1$                                       | $\theta_2$ | $\theta_3$ |
| Metal structures                                                     | 80                                               | 6          | 4          |
| Reinforced concrete, concrete, reinforced stone and stone structures |                                                  |            |            |
| less than 15 cm thick                                                | 8                                                | 6          | 4          |
| from 15 to 39 cm thick                                               | 6                                                | 4          | 6          |
| more than 40 cm thick                                                | 2                                                | 2          | 4          |

Notes: 1. In heated industrial buildings at the stage of use it is possible not to consider  $\Delta_7$  for the structures defended against the solar radiation.

2. In highlands and insufficiently explored regions of the USSR shown on the maps 5 – 7 of the obligatory Annex 5  $t_{ew}$ ,  $t_{ec}$  are calculated by the following formulas:

$$t_{ec} = t_{1,\min} + 0.5A_1 \quad (19)$$

$$t_{ew} = t_{VII,\max} - 0.5A_{VII} \quad (20)$$

where  $t_{1,\min}$ ,  $t_{VII,\max}$  – are average of absolute values correspondingly of minimum temperature in January and maximum temperature in July;

$A_1$ ,  $A_{VII}$  – average daily ranges of air temperature correspondingly in January and July by clear sky.

$t_{1,\min}$ ,  $t_{VII,\max}$ ,  $A_1$ ,  $A_{VII}$  are taken according to the State hydrometeorology committee.

**8.5.** Increment  $\theta_4$  and  $\theta_5$  in degree Celsius must be calculated by the following formulas

$$\theta_4 = 0.05\rho S_{\max}kk_1 \quad (21)$$

$$\theta_5 = 0.05\rho S_{\max}k(1 - k_1) \quad (22)$$

where  $\rho$  – coefficient of solar radiation absorption by the material of the outside surface of the structure, taken in compliance with SNiP II-3-79\*;

$S_{\max}$  – maximum value of total solar radiation (direct and scattered), watt per square meter taken in compliance with SNiP 2.01.01-82;

$k$  – coefficient taken according to the table 17;

$k_1$  – coefficient taken according to the table 18.

Table 17

| Kind and orientation of the surface (surfaces) | Coefficient $k$ |
|------------------------------------------------|-----------------|
| Horizontal                                     | 1.0             |
| Vertical, oriented to the:                     |                 |
| south                                          | 1.0             |
| west                                           | 0.9             |
| east                                           | 0.7             |

Table 18

| Structures of the building                                           | Coefficient $k_1$ |
|----------------------------------------------------------------------|-------------------|
| Metal structures                                                     | 0.7               |
| Reinforced concrete, concrete, reinforced stone and stone structures | 0.6               |
| less than 15 cm thick                                                | 0.4               |
| from 15 to 39 cm thick                                               | 0.3               |
| more than 40 cm thick                                                |                   |

**8.6.** Initial temperature which conforms to the locking of the structure or its part into the completed system during warm  $t_{0w}$  and cold  $t_{0c}$  season must be calculated by the following formulas

$$t_{0w} = 0.8t_{VII} + 0.2t_1 \quad (23)$$

$$t_{0c} = 0.2t_{VII} + 0.8t_1 \quad (24)$$

## 10. Deflections and Transitions

The norms of the present chapter fix the limit deflections and transitions of load-carrying and walling structures of the buildings by the calculation as regards the second group of limit states independently on the building materials.

The norms do not apply to the hydraulic works, transport facilities, nuclear power-stations as well as to power line supports, open distribution devices and antenna communication constructions.

### General Instructions

**10.1.** During calculation of the structures as regards the deflections and transitions it is necessary to meet the following condition

$$f \leq f_u \quad (25)$$

where  $f$  – deflection and transition of the member of the structure (or the whole structure) calculated considering the factors which influence on the their value in compliance with items 1 – 3 of the recommended Annex 6.

$f_u$  – is limit deflection and transition fixed by the present norms.

The calculation must be made according to the following requirements:

- a) technological (provide technological and lifting equipment, control instrumentation service conditions);
- b) constructive (provide connection integrity of structures, their members and joints, provide necessary inclining);
- c) physiological (prevent deleterious effects, discomfort sense during vibrations);
- d) aesthetic and psychological (provide beneficial impressions by outward appearance of structures, prevent danger sense).

Each of mentioned above requirements must be completed by calculation independently on the others.

The limitation of the vibrations of the structure must be fixed in compliance with normative documents of item 4 of the recommended Annex 6.

**10.2.** The design situations for which it is necessary to calculate the deflections and transitions, corresponding to them loads as well as the requirements concerning the building raising are given in item 5 of the recommended Annex 6.

**10.3.** Roof and floor structures members Limit deflections confined according to the technological, constructive and physiological requirements must be counted out of the bent axis from which the deflection is calculated and which corresponds to the state of the member at the moment of the load action; the deflections confined according to the aesthetic and psychological requirements are counted out of the straight line which connects the supports of these members (see also item 7 of the recommended Annex 6).

**10.4.** The members of the structures deflections are not confined according to the aesthetic and psychological requirements if they do not worsen the appearance of the structure (for example thin-shell roof, sloped sun shields, structures with weighed down or uplifted bottom chord) and if the members of the structure are concealed. The deflections are not confined in the buildings where people stay short-term period (for example transformer substations, attics).

Note. For all kinds of the floor structures the continuity of the roof covering must be provided by means of constructive measures (for example using compensators) but not by means of the increasing of the durability of the structure.

- 10.5.** Safety factor of all considered loads and the dynamic coefficient of the loads from the loaders, electric carts, bridge cranes and overhead cranes must be taken as equal to one.

Safety factor according to the responsibility must be taken in compliance with the obligatory Annex 7.

- 10.6.** For the members of the structures or buildings limit deflections and transitions which are not stipulated in the present or other normative documents vertical and horizontal deflections caused by dead loads, long-term loads and short-term loads must not exceed  $1/150$  part of the bay or  $1/75$  part of the console overhang.

#### **Vertical limit deflections of the structure members**

- 10.7** Vertical limit deflections of the structures members and loads from which it is necessary to calculate the deflections are given in item 6 of the recommended Annex 6.
- 10.8.** The distance (the gap) from the top point of the bridge crane crab to the bottom point of deflected load-carrying structures of the roof (or objects connected to them) must be no less than 100 mm.
- 10.9.** The deflections of the roof members must be so as to provide the slope of roof no less than  $1/200$  in one direction (except the cases stipulated in other normative documents).

Table 19

| Members of the structure                                                                                                                  | Requirements                    | Vertical limit deflections $f_u$                           | Load for the determination of vertical deflections                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Gantry rail beams for bridge cranes and overhead crane, operated:<br>from the floor, including telfers, (chain blocks)                 | Technological                   | $l/25$                                                     | From one crane                                                                                                                                |
| from the crane cabin by the operating mode groups (according to the GOST 25546-82):                                                       | Physiological and technological |                                                            |                                                                                                                                               |
| 1K – 6K                                                                                                                                   |                                 | $l/400$                                                    | The same                                                                                                                                      |
| 7K                                                                                                                                        |                                 | $l/500$                                                    | The same                                                                                                                                      |
| 8K                                                                                                                                        |                                 | $l/600$                                                    | The same                                                                                                                                      |
| 2. Beams, frameworks, collar beams, slabs, coverings (including cross ribs of the slabs and coverings):                                   |                                 |                                                            |                                                                                                                                               |
| a) of floors and roofs which are open for the observing if the bay $l$<br>$l \leq 1$<br>$l=3$<br>$l=6$<br>$l=24$ (12)<br>$l \geq 36$ (24) | Aesthetic and psychological     | $l/120$<br>$l/150$<br>$l/200$<br>$l/250$<br>$l/300$        | Dead loads and temporary sustained load                                                                                                       |
| b) of the roof or of the floor if there are walls under it                                                                                | Constructive                    | Taken in compliance with item 6 of the recommended Annex 6 | The loads which cause the decreasing of the gap between the load-carrying members of the structures and the walls, located under the members. |
| c) of the roof and of the floor if there are members on them which are subjected to cracks (bracing wires, walls)                         | Constructive                    | $l/150$                                                    | The loads acting after the walls, floors and [] are made.                                                                                     |
| d) of the roofs and of the floors if there are telfers (chain blocks), overhead cranes which are operated                                 |                                 |                                                            |                                                                                                                                               |
| from the floor                                                                                                                            | Technological                   | $l/300$ or $a/150$<br>(the less one of two)                | Temporary load considering the load from one crane or a telfer (chain block) on one rail.                                                     |



Table 19

| Members of the structure                                                                                                                  | Requirements                    | Vertical limit deflections $f_u$                           | Load for the determination of vertical deflections                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Gantry rail beams for bridge cranes and overhead crane, operated:<br>from the floor, including telfers, (chain blocks)                 | Technological                   | $l/25$                                                     | From one crane                                                                                                                                |
| from the crane cabin by the operating mode groups (according to the GOST 25546-82):                                                       | Physiological and technological |                                                            |                                                                                                                                               |
| 1K – 6K                                                                                                                                   |                                 | $l/400$                                                    | The same                                                                                                                                      |
| 7K                                                                                                                                        |                                 | $l/500$                                                    | The same                                                                                                                                      |
| 8K                                                                                                                                        |                                 | $l/600$                                                    | The same                                                                                                                                      |
| 2. Beams, frameworks, collar beams, slabs, coverings (including cross ribs of the slabs and coverings):                                   |                                 |                                                            |                                                                                                                                               |
| e) of floors and roofs which are open for the observing if the bay $l$<br>$l \leq 1$<br>$l=3$<br>$l=6$<br>$l=24$ (12)<br>$l \geq 36$ (24) | Aesthetic and psychological     | $l/120$<br>$l/150$<br>$l/200$<br>$l/250$<br>$l/300$        | Dead loads and temporary sustained load                                                                                                       |
| f) of the roof or of the floor if there are walls under it                                                                                | Constructive                    | Taken in compliance with item 6 of the recommended Annex 6 | The loads which cause the decreasing of the gap between the load-carrying members of the structures and the walls, located under the members. |
| g) of the roof and of the floor if there are members on them which are subjected to cracks (bracing wires, walls)                         | Constructive                    | $l/150$                                                    | The loads acting after the walls, floors and [] are made.                                                                                     |
| h) of the roofs and of the floors if there are telfers (chain blocks), overhead cranes which are operated                                 |                                 |                                                            |                                                                                                                                               |
| from the floor                                                                                                                            | Technological                   | $l/300$ or $a/150$<br>(the less one of two)                | Temporary load considering the load from one crane or a telfer (chain block) on one rail.                                                     |

| Members of the structure                                                                                         | Requirements                    | Vertical limit deflections $f_u$             | Load for the determination of vertical deflections                                                                                     |
|------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| from the cabin                                                                                                   | Physiological                   | $l/400$ or $a/200$<br>(the less one of two)  | The load from one crane or telfer (chain block) on one rail                                                                            |
| i) of the floors exposed to the effect of:                                                                       | Physiological and technological |                                              |                                                                                                                                        |
| moved cargos, materials, joints and equipment elements and other movable loads (including by the road transport) |                                 | $l/350$                                      | 0.7 of total standard values of temporary loads or of the load from one loader (the most adverse of two)                               |
| the loads from rail transport<br>narrow-gauge<br><br>broad-gauge                                                 |                                 | $l/400$<br><br>$l/500$                       | The load from one train on one rail<br>The same                                                                                        |
| 3. The members of the stairs (flights of the stairs, stair platforms, strings), balconies, loggias               | Aesthetic and psychological     | The same like in Position 2, $a$             |                                                                                                                                        |
|                                                                                                                  | Physiological                   | Are determined in compliance with item 10.10 |                                                                                                                                        |
| 4. Floor slabs, stair platforms and flights of the stairs, to whose deflection the adjoining members oppose.     | Physiological                   | 0.7 mm                                       | Point load 1 kN (100 kilogram-force) in the middle of the                                                                              |
| 5. Cross pieces and hinged plates above the window and door openings                                             | Constructive                    | $l/200$                                      | The loads that cause the decreasing of the gap between the load-carrying members and window or door filling, located under the members |
|                                                                                                                  | Aesthetic and psychological     | The same like in position 2, $a$             |                                                                                                                                        |

Symbols taken in table 19:

$l$  – design span of the structure member;

$a$  – beams or truss spacing to which the suspended crane rails are fixed

Notes: 1. For a console it is necessary to take its double overhang instead of  $l$ .

2. For middle values  $l$  in position 2,  $a$  the limit deflections must be calculated by means of linear interpolation considering the item 7 requirements of the recommended Annex 6.

3. In the position 2,  $a$  the numbers shown in brackets must be taken if the height of the rooms is up to 6 meters inclusive.

4. Peculiarity of the deflections calculation according to the position 2,  $d$  are shown in item 8 of the recommended Annex 6.

5. If the deflections are limited by aesthetic and psychological requirements so it is possible to take the span  $l$  as equal to the distance between the inside surfaces of the load-carrying walls (or columns).

**10.10.** The limit deflections of the floor members (beams, slabs), of the stairs, balconies, loggias, limit deflections of the rooms of apartment buildings and office blocks as well as limit deflections of amenity rooms of industrial buildings based on physiological requirements must be calculated by the following formula

$$f_u = \frac{g(p + p_1 + q)}{30n^2(bp + p_1 + q)} \quad (26)$$

where  $g$  – acceleration of gravity

$p$  – standard load from people who cause the vibration, taken according to the table 20

$p_1$  – decreased standard value of the load upon the floor slabs, taken according to tables 3 and 20

$q$  – standard value of the load from the weight of the calculated member and the members bearing against it

$n$  – frequency of the load during the walking of a person, taken according to table 20

$b$  – coefficient taken according to table 20

| Rooms, taken according to table 3                                                                | $\rho$ , kPa (kilogram-force per square meter) | $\rho_1$ , kPa (kilogram-force per square meter) | $n$ , Hertz | $b$                              |
|--------------------------------------------------------------------------------------------------|------------------------------------------------|--------------------------------------------------|-------------|----------------------------------|
| Pos 1, 2, except class rooms and amenity rooms;<br>Pos 3, 4a, 9b, 10b                            | 0.25 (25)                                      | Taken according to table 3                       | 1.5         | $125\sqrt{\frac{Q}{\alpha p l}}$ |
| Pos 2 – class rooms and amenity rooms;<br>Pos 4b – g, except dance rooms;<br>Pos 9a, 10a, 12, 13 | 0.5 (50)                                       | The same                                         | 1.5         | $125\sqrt{\frac{Q}{\alpha p l}}$ |
| Pos 4 – dance rooms;<br>Pos 6, 7                                                                 | 1.5 (150)                                      | 0.2 (20)                                         | 2.0         | 50                               |

*Symbols taken in Table 20:*

$Q$  – the weight of one person taken as equal to 0.8 kN (80 kilogram-force);

$\alpha$  – coefficient taken as equal to 1.0 for the members calculated as regards the beam lay-out, and taken as equal to 0.5 – in other cases (for example if the slab is supported by three or four sides)

$a$  – spacing of the beams, collar beams, the width of the slabs (coverings), in meters

$l$  – design span of the structure member; in meters

The deflections must be calculated as regards the sum of the loads  $\psi_{A1}p + p_1 + q$  where  $\psi_{A1}$  – is the coefficient calculated as regards formula (1).

# SNiP II-23-81

## CONCRETE AND REINFORCED CONCRETE STRUCTURES

### 5. Constructive Requirements

**Section 5.1.** By the designing of concrete and reinforced concrete structures to provide the conditions of their manufacturing, required durability and combined work of reinforcement and concrete it is necessary to follow the constructive requirements given in the present part.

#### Section of the Member Minimum Dimensions

**Section 5.2.** Minimum dimensions of concrete and reinforced concrete members sections calculated on base of acting forces and corresponding groups of limiting states, must be fixed considering economical requirements, necessity of forms and reinforcement unification as well as the conditions of fabrication method of structures.

Besides, the dimensions of reinforced concrete structures sections must be taken so as to follow the requirements concerning the location of reinforcement in the section (thickness of protective layers of concrete, distances between iron bars, etc.) and anchorage of reinforcement.

**Section 5.3.** The thickness of monolithic slabs must be taken no less than:

|                                                                                  |    |
|----------------------------------------------------------------------------------|----|
| for roofing .....                                                                | 40 |
| for floor of living and public buildings.....                                    | 50 |
| for floor of industrial buildings.....                                           | 60 |
| for slabs made of light-weight concrete of class B7.5 and lower in any case..... | 70 |

Minimum thickness of prefabricated slabs must be calculated according to the condition of the providing of concrete protective layer and conditions of reinforcement location along the slab thickness (see items 5.4 – 5.12).

The dimensions of eccentric pressed members section must be taken so as their flexibility  $l_0 / i$  in any direction didn't exceed the following values:

|                                                                                           |     |
|-------------------------------------------------------------------------------------------|-----|
| for reinforced concrete members made of heavy-weight, fine and light-weight concrete..... | 200 |
| for columns which are the members of the buildings.....                                   | 120 |
| for concrete members made of heavy-weight, light-weight and porous concrete.....          | 90  |
| for concrete and reinforced concrete members made of cellular concrete.....               | 70  |

#### Protective Layer of Concrete

**Section 5.4.** The protective layer of concrete for working reinforcement must provide combined work of reinforcement with concrete on all structure work stages; it must also provide the protection of reinforcement from outside atmospheric, temperature and other forcing.

**Section 5.5.** For longitudinal working reinforcement (stressed and no stressed, stretched on the supports) the thickness of the protective layer must be as a rule no more than the diameter of a bar or of a rope and no less than (in millimeters)

|                                                                   |    |
|-------------------------------------------------------------------|----|
| in slabs and walls which are 100 and more millimeters thick.....  | 10 |
| in slabs and walls which are less than 100 millimeters thick..... | 15 |
| in beams and ribs which are less than                             |    |

250 millimeters high.....15

|                                                                      |    |
|----------------------------------------------------------------------|----|
| in beams and ribs which are 250 and more millimeters thick.....      | 20 |
| in columns.....                                                      | 20 |
| in foundation beams.....                                             | 30 |
| in prefabricated foundations.....                                    | 30 |
| in monolithic foundations with concrete structural supporting.....   | 35 |
| in monolithic foundation without concrete structural supporting..... | 70 |

In one-layer structures made of light-weight and porous concrete of B7.5 and lower class the thickness of the protective layer must be no less than 20 millimeters; for outside wall slabs (without face finishing layer) it must be no less than 25 millimeters.

In one-layer structures made of cellular concrete the thickness of the protective layer must be no less than 25 millimeters in any case.

**Section 5.6.** The thickness of the concrete protective layer for lateral, distribution and constructive reinforcement must be taken no less than the diameter of the mentioned reinforcement and no less than:

10 millimeters if the section of the member is less than 250 millimeters;  
15 millimeters if the section of the member is 250 millimeters and more

In the members made of light and porous concrete of B7.5 class and lower, of cellular concrete independently on the section height the thickness of the protective layer of the concrete for the crossbar is no less than 15 millimeters.

### **Minimum distance between the reinforcing rods**

**Section 5.11.** The distance between the iron bars (or canal casings) along the height and width of the section must provide reinforcement and concrete combined work and must be determined considering the convenience of packing of concrete; for pre-stressed structures local reduction degree and dimensions of stretcher (a lifting jack, cramps etc.) must be also taken into account. In the members which are made by means of vibrostamp machine or bayonet vibrator the elements of these machines which provide the concrete consolidation must have free way between the iron bars of the member.

**Section 5.12.** The distances between the separate rods of the longitudinal non pre-stressed or pre-stressed reinforcement strained on the bearings as well as between the longitudinal reinforcement rods of the neighbor plane welded frameworks must be taken no less than maximum rod diameter and also:

- a) if the rods during the concreting take the horizontal or inclined position, so this distance must be no less than 25 millimeters for bottom reinforcement and no less than 30 millimeters for top reinforcement; if the bottom reinforcement is located in more than two rows so the distance between the iron bars in horizontal direction (except the iron bars of two bottom rows) must be no less than 50 millimeters;
- b) if the bars take vertical position during the concreting – no less than 50 millimeters; by the systematic concrete aggregate fractionating control this distance can be decreased up to 35 millimeters but at the same time this distance must be no less than maximum dimension of coarse aggregate multiplied by 1.5.

By tight conditions it is possible to place the iron bars in pairs (without any gap between them).

In the members with pre-stressed reinforcement which is stretched on the concrete (with the exception of continuously reinforced structures) the clear spacing between the reinforcement canals must be as a rule no less than the diameter of the canal, at any case no less than 50 millimeters.

Note. The clear spacing between the iron bars of die-rolled section is taken according to nominal diameter without considering overshoots and arrises.

## **SNIP 2.02.04-88 Construction Codes and Regulations**

requirements to be observed during laying of communication lines inside the built-up territories.

### **FOUNDATION DEPTHS**

3.7. Foundation depth measured from the level of grading (filling, grading) shall be assigned pursuant to the requirements of SNIP 2.02.01-83, the adopted concept of permafrost employment as base and should be confirmed by calculation of foundation stability against the forces of ground frost heaving (as per Paras 4.41 and 4.45).

3.8. When Concept I is applied to permafrost ground use as base, the minimal foundation depth  $d_{min}$  should be selected according to Table 1 with regard to the estimated depth of ground seasonal thawing  $d_{th}$  determined by use of the mandatory Appendix 3.

Table 1

| Foundation Type                                               | Minimal Foundation Depth<br>$d_{min}$ , m |
|---------------------------------------------------------------|-------------------------------------------|
| All types, except piles foundations                           | $d + l$                                   |
| Piles foundations of buildings and structures                 | $d + h + 2$                               |
| Bridge pier piles                                             | $d + 4$                                   |
| Foundations of buildings and structures erected upon fillings | I<br>Non-normalized                       |

3.9. When Concept II is applied to permafrost ground use as base, the minimal foundation depth  $d$  should be selected according to the requirements of SNIP 2.02.01-83 with regard to the estimated depth of ground seasonal freezing  $d_f$  determined by use of the mandatory Appendix 3, and the level of ground waters adopted with regard to the development of a ground defrosting zone beneath the structure.

Laying of foundations shall not be allowed inside seasonal freezing/thawing beds unless substantiated by adequate calculation of bases and foundations (Para 4.45).

## CONSTRUCTION IN SEISMIC REGIONS SNIP II-7-81

- 1.1** The present Building Codes and Rules shall apply to designing of buildings and structures in areas with a seismicity of 7, 8 and 9.
- 1.2** In designing buildings and structures in areas with the above seismicity, one shall:
- use materials, structural elements and structural concepts ensuring the lowest possible seismic loads
  - base structural design on principles of symmetry, even distribution of stiffness and mass of structural elements, as well as even distribution of loads on floors;
  - remove joints in prefabricated buildings and structures from the zone of maximum forces and make the buildings and structures as monolithic and uniform as possible by using large prefabricated units;
  - create conditions conducive to plastic strain in structural elements and joints therebetween and at the same time ensure structural stability.

To ensure seismic resistance of buildings and structures, their seismic isolation and other systems for controlling their dynamic response may be used, provided such buildings and structures are designed as per project specific requirements approved by the RF State Committee for Construction

Table 1\*

| Seismic soil category | Soils                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Site seismicity at the following seismicity of the area: |   |     |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|---|-----|
|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7                                                        | 8 | 9   |
| I                     | Rocky soils of all types (including permafrost and thawed permafrost), unweathered and slightly weathered, coarsely clastic, compact, with low moisture content, igneous, containing up to 30 % of sandy and clayey filler; weathered and heavily weathered, rocky and non-rocky frozen (permafrost) at a temperature of -2° C and lower in the case of construction and operation based on the principle that the soil at the foundation is maintained frozen.                                                                                                                                                                  | 6                                                        | 7 | 8   |
| II                    | Rocky soils, weathered and heavily weathered, including permafrost, with the exception of category I soils; gravelly sands, coarse and medium, compact and medium, with low to medium moisture content; fine and silty sands, compact and medium, with low moisture content; clayey soils with consistency index $I_L \leq 0.5$ at porosity coefficient $e < 0.9$ for clays and loams and $e < 0.7$ for sandy loams; perpetually frozen non-rocky soils, plastic- or loose-frozen, as well as permafrost at a temperature in excess of -2° C in the case of construction and operation based on the principle as per category I. | 7                                                        | 8 | 9   |
| III                   | Loose sands irrespective of moisture content and particle size; gravelly sands, coarse and medium, compact and medium, water-saturated; fine and silty sands, compact and medium, moist and water-saturated; clayey soils with consistency index $I_L > 0.5$ ; clayey soils with consistency index $I_L \leq 0.5$ at porosity coefficient $e \geq 0.9$ for clays and loams and $e \geq 0.7$ for sandy loams; perpetually frozen non-rocky soils in the case of construction and operation based on the principle as per category II (thawing of the soil at the foundation is allowed).                                          | 8                                                        | 9 | > 9 |

Notes:

1\*. Seismic category I may be assigned to a construction site if the layer corresponding to category I is more than 30 m thick from the grade.

If the soil is non-homogeneous, the less favorable seismic category shall be assigned if within a 10 m thick layer (from the grade) the total thickness of the layer belonging to this category is in excess of 5 m.

2. When the level of groundwater rise and the waterlogging of soils (including those prone to subsidence) in the course of operation of a building or structure are predicted, the soil category shall be determined depending on the soil properties (moisture content, consistency) in the wetted state.

3. In the case of construction on perpetually frozen non-rocky soils based on the principle as per category II, if the thawing zone reaches the underlying thawed soil, the soil at the foundation shall be regarded as non-permafrost (in accordance with its actual state after thawing).

4. In the case of buildings and structures of special importance, erected in areas with a seismicity of 6 on sites with soils of seismic category III, the design seismicity shall be taken equal to 7.

5. The seismicity of construction sites for transport and hydraulic structures shall be determined with due account for the additional requirements set forth in Sections 4 and 5.

6. In the absence of consistency or moisture content data, seismic category III shall be assigned to clayey and sandy soils with a groundwater level higher than 5 m.

## 2. DESIGN LOADS

**2.1** The structural elements and foundations of buildings and structures designed for erection in seismic areas shall be designed to main loads and special load combinations with due account for seismic forces.

When buildings and structures are designed to special load combinations, the design load values shall be multiplied by the combination coefficients listed in Table 2.

Table 2

| Load types                           | Value of combination coefficient $n_c$ |
|--------------------------------------|----------------------------------------|
| Permanent                            | 0.9                                    |
| Temporary sustained                  | 0.8                                    |
| Short-duration (on floors and roofs) | 0.5                                    |

In this case, horizontal loads due to masses on flexible suspensions, temperature and climatic effects, wind loads, dynamic loads from equipment and vehicles, braking and lateral forces associated with crane movements shall not be taken into account.

When the design vertical seismic load is determined, the weights of the crane bridge, crane trolley and crane load equal to the crane's rated load capacity multiplied by 0.3 shall be factored in.

The design horizontal seismic load due to crane bridge weight shall be taken in a direction normal to the axis of the crane girder. In this case, the lowering of crane loads, stipulated in SNiP covering loads and forces, shall not be taken into account.

**2.2** In the case of buildings and structures designed to special load combinations with due account for seismic forces, the design shall include:

- (a) loads determined as per paragraph 2.5 hereinbelow;
- (b) measured accelerations of the foundation during earthquakes particularly hazardous for a given building or structure, as well as synthetic accelerograms; in this case, the maximum amplitudes of foundation accelerations shall be taken equal to at least 100, 200 or 400 cm/sq. s at a site seismicity of 7, 8 and 9, respectively.

In calculations as per (b), the possibility of inelastic strain of structural elements shall be taken into account.

Calculations as per (a) shall be made for all buildings and structures.



Calculations as per (b) shall be made for buildings of special importance and high-rises (higher than 16 storeys).

### 2.3 Seismic forces may have any spatial orientation.

In the case of buildings and structures of simple geometry, the design seismic loads shall be assumed to be directed horizontally with respect to their longitudinal and transverse axes. The seismic loads in these directions shall be taken into account separately.

In the case of structures of complex geometry, seismic loads acting in the most hazardous directions for a given structure or its components shall be taken into account.

### 2.4 The vertical seismic load shall be taken into account when designing:

- horizontal and inclined cantilever structures;
- bridge spans structures;
- frameworks, arches, girders and shell roofs with a span of 24 m and more;
- structures with inherent stability toward tipping or slipping forces;
- masonry structures (as per paragraph 3.37 hereinbelow).

### 2.5 Design seismic load $S_{ik}$ in a given direction, applied at point $k$ and corresponding to the $i$ th mode of the natural vibration of buildings and structures (with the exception of hydraulic structures) shall be determined using the formula:

$$S_{ik} = K_1 S_{0oik}, \quad (1)$$

where  $K_1$  is a coefficient taking into account tolerable damage to buildings and structures, taken from Table 3\*;

$S_{0oik}$  is the seismic load for the  $i$ th mode of the natural vibration of a building or structure, determined with the assumption of elastic strain in structural elements from the formula

$$S_{0oik} = Q_k A \beta_i K_\psi \eta_{ik}, \quad (2)$$

where  $Q_k$  is the weight of the building or structure at point  $k$ , determined taking into account the design loads on the structural element as per paragraph 2.1 hereinabove (Fig. 1);

$A$  is a coefficient equal to 0.1, 0.2 or 0.4, respectively, for the design seismicity of 7, 8 or 9;

$\beta_i$  is the impact factor corresponding to the  $i$ th mode of the natural vibration of a building or structure, determined as per paragraph 2.6\* hereinbelow;

$K_\psi$  is a coefficient taken from Table 6\* or as per Section 5 hereinbelow;

$\eta_{ik}$  is a coefficient dependent on the strain in a building or structure, corresponding to the  $i$ th mode of its natural vibration, and also on the load application point, determined as per paragraph 2.7 hereinbelow.

Note: Design seismicity for building or structures, as well as values of coefficient  $K_1$  shall be taken from Tables 3 and 5, subject to approval by the organization accepting the design.

In the case of a construction site seismicity of 8 and higher and soils of category III, a factor of 0.7 shall be applied to  $S_{ik}$ , taking into account the nonlinear strain of soils during seismic events.

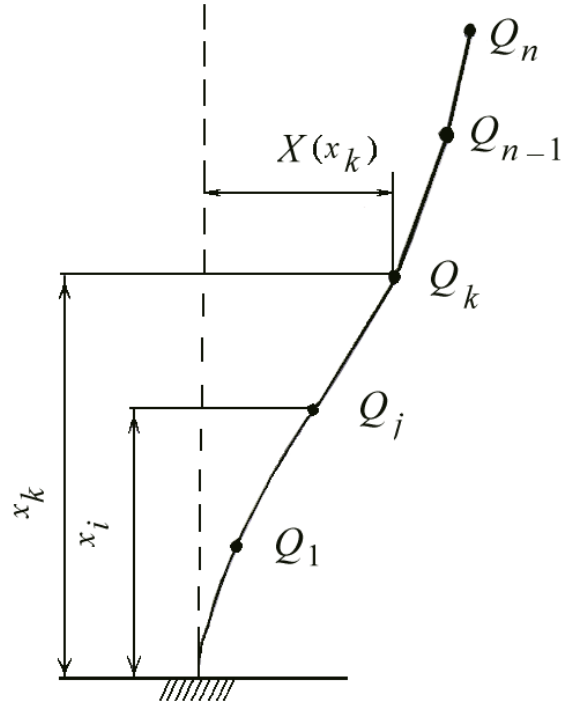


Fig. 1

**2.6** Depending on the calculated period  $T_i$  of the  $i$ th mode natural vibration of a building or structure, impact factor  $\beta_i$  for determination of seismic loads shall be defined using formulas (3) and (4) or Figure 2.

For soils of seismic categories I and II (curve 1)

$$\begin{aligned} \text{at } T_i \leq 0.1 \text{ s} & \quad \beta_i = 1 + 15T_i \\ \text{at } 0.1 \text{ s} < T_i < 0.4 \text{ s} & \quad \beta_i = 2.5 \\ \text{at } T_i \geq 0.4 \text{ s} & \quad \beta_i = 2.5(0.4/T_i)^{0.5} \end{aligned} \quad (3)$$

For soils of seismic category III (curve 2)

$$\begin{aligned} \text{at } T_i \leq 0.1 \text{ s} & \quad \beta_i = 1 + 15T_i \\ \text{at } 0.1 \text{ s} < T_i < 0.8 \text{ s} & \quad \beta_i = 2.5 \\ \text{at } T_i \geq 0.8 \text{ s} & \quad \beta_i = 2.5(0.8/T_i)^{0.5} \end{aligned} \quad (4)$$

In any event, impact factor  $\beta_i$  shall be taken equal to at least 0.8.

Note: In the case of transport and hydraulic structures, relations  $\beta_i(T)$  shall be selected as per Sections 4 and 5 hereinbelow.

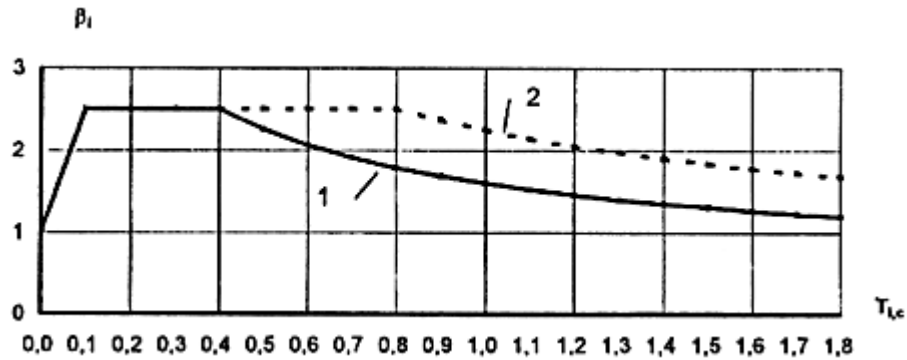


Fig. 2

**2.7** In the case of buildings and structures based on a cantilever design, coefficient  $\eta_{ik}$  shall be determined using the formula:

$$\eta_{ik} = \frac{X_i(x_k) \sum_{j=1}^n Q_j X_i(x_j)}{\sum_{j=1}^n Q_j X_i^2(x_j)}, \quad (6)$$

where  $X_i(x_k)$  and  $X_i(x_j)$  are displacements of a building or structure, corresponding to the  $i$ th mode of their natural vibration, at point  $k$  and at all points  $j$  where its weight is assumed to be concentrated in accordance with the calculations;

$Q_j$  is the weight of a building or structure at point  $j$ , determined taking into account the design loads on the building or structure as per paragraph 2.1 hereinabove.

**2.8** In the case of buildings with up to five storeys inclusive and with insignificant mass and rigidity variations from one floor to another, at  $T_1 < 0.4$ , coefficient  $\eta_{ik}$  may be determined using the following simplified formula:

$$\eta_{ik} = \frac{x_k \sum_{j=1}^n Q_j x_j}{\sum_{j=1}^n Q_j x_j^2}, \quad (7)$$

where  $x_k$  and  $x_j$  are the distances from points  $k$  and  $j$  to the top edge of the foundations.

**2.9** The forces arising in buildings and structures designed for seismic areas shall be determined taking into account at least three modes of natural vibration if the period of the first (lowest) natural vibration mode,  $T_1$ , exceeds 0.4 s and only the first mode if  $T_1$  is equal to or less than 0.4 s.

The number of modes and coefficients  $\eta_{ik}$  for hydraulic structures shall be selected as per Section 5 hereinbelow.

**2.10** The design values of the transverse and longitudinal forces, bending and tilting moments, as well as normal and tangential stresses  $N_p$  in structural elements due to a static seismic load shall be determined from the formula:

$$N_p = \sqrt{\sum_{i=1}^n N_i^2}, \quad (8)$$

where  $N_i$  stands for forces or stresses across the section under analysis, caused by seismic loads corresponding to the  $i$ th mode;

$n$  is the number of modes used in the calculation.

**2.11** In the cases listed in paragraph 2.4 hereinabove (with the exception of masonry structures), the vertical seismic load shall be determined using formulas (1) and (2) with coefficients  $K_\psi$  and  $K_2$  being taken equal to unity.

Cantilever members whose weight is insignificant as compared to that of the building (balconies, canopies, cantilevers for hanging walls, and so on, as well as their fastening systems) shall be designed for seismic loading at  $\beta_\eta = 5$ .

**2.12** Structural elements rising above a building or structure and having comparatively small cross-sections and weights (parapets, pediments, etc.), as well as anchoring of heavy sculptures or equipment installed on the ground floor shall be designed taking into account the horizontal seismic load calculated using formulas (1) and (2) at a  $\beta_\eta = 5$ .

**2.13** Walls, panels, partitions, connections between individual structural elements and process equipment anchoring shall be designed taking into account the horizontal seismic load calculated using formulas (1) and (2) at a  $\beta_\eta$  value corresponding to the level at which they are installed in the building or structure, but not less than 2.

Forces of friction shall be taken into account only in the design of horizontal abutting joints in large-panel buildings.

**2.14** When structural elements are designed for strength and stability, in addition to the operating conditions coefficients adopted in accordance with other codes and rules (SNIIP) of part II, additional working conditions coefficient  $m_{wc}$  shall be introduced, taken from Table 7.

**2.15** When buildings and structures (with the exception of hydraulic structures) longer or wider than 30 m are designed, in addition to the seismic load determined as per paragraph 2.5 hereinabove, one shall also take into account the torque relative to the vertical axis of the building or structure, passing through its center of rigidity. The value of the design eccentricity between the centers of rigidity and mass of buildings or structures at the level under analysis shall be taken equal to at least  $0.1B$ ,  $B$  being the dimension of the building or structure in plan view in a direction normal to load  $S_{ik}$ .

**2.16** Retaining walls shall be designed taking into account the seismic soil pressure.

**2.17** The design of buildings and structures with due account for seismic forces shall, as a rule, be based on first-group limit states. In cases stipulated by process requirements, the design may be based on second-group limit states.

Table 3

| Building or structure type                                                                                                                                                                                                                  | $K_1$ |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1. Buildings and structures in which no damage or inelastic strains are allowed.                                                                                                                                                            | 1     |
| 2. Buildings and structures whose structural elements may undergo residual strains and damage hindering normal operation but without affecting personnel and equipment, made up of:                                                         |       |
| large panel or cast-in-place reinforced concrete members                                                                                                                                                                                    | 0.22  |
| a steel frame without vertical diaphragms or bracing                                                                                                                                                                                        | 0.25  |
| ditto, with vertical diaphragms or bracing                                                                                                                                                                                                  | 0.22  |
| a reinforced concrete frame without vertical diaphragms or bracing                                                                                                                                                                          | 0.35  |
| ditto, with vertical diaphragms or bracing                                                                                                                                                                                                  | 0.25  |
| brickwork or masonry                                                                                                                                                                                                                        | 0.35  |
| 3. Buildings and structures whose structural elements may undergo substantial residual strains, cracking, damage of individual members and their displacement, temporarily bringing normal operation to a halt without affecting personnel. | 0.12  |

**2.18** The need to take seismic forces into account when designing buildings and structures whose destruction does now cause human mortality, damage to valuable equipment and interruption of continuous production processes (warehouses, crane trestles, small workshops, etc.), as well as temporary buildings and structures, shall be established by the customer.

Table 6

| Building or structure type                                                                                                             | $K_{\psi}$ |
|----------------------------------------------------------------------------------------------------------------------------------------|------------|
| 1. Tall structures with small footprint (towers, masts, smoke stacks, separate elevator shafts, etc.)                                  | 1.5        |
| 2. Buildings with stanchions on the ground floor, the strain capacity ratio between the first and ground floors being 0.25 and greater |            |
| 3. Skeleton-type buildings whose walls do not affect their strain capacity                                                             |            |
| 3. Buildings and structures not covered by Items 1 and 2 hereinabove, with the exception of hydraulic structures                       | 1          |

Table 7

| Structure type                                                                                                      | $m_{wc}$                         |
|---------------------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>In strength analyses</b>                                                                                         |                                  |
| 1. Steel, wooden, reinforced concrete with stiff reinforcement                                                      | 1.3                              |
| 2. Reinforced concrete with bar and wire reinforcement, with the exception of strength analysis of oblique sections | 1.2                              |
| 3. Reinforced concrete, strength analysis of oblique sections                                                       | 1.0                              |
| 4. Masonry, reinforced brickwork or masonry, concrete eccentric compression strength analysis                       | 1.0                              |
| shear and tensile strength analysis                                                                                 | 0.8                              |
| 5. Welded joints                                                                                                    | 1.0                              |
| 6. Bolted and riveted connections                                                                                   | 1.1                              |
| <b>In stability analyses</b>                                                                                        |                                  |
| 7. Steel members with a ductility above 100                                                                         | 1.0                              |
| 8. Ditto, with a ductility of up to 20                                                                              | 1.2                              |
| 9. Ditto, with a ductility of 20 to 100                                                                             | from 1.2 to 1.0 by interpolation |

Note: In strength analyses of steel and reinforced concrete members designed for use in unheated premises or outdoors at a design temperature below  $-40^{\circ}\text{C}$ ,  $m_{wc}$  should be taken equal to 0.9, and in strength analyses involving oblique sections,  $m_{wc}$  should be taken equal to 0.8.

### 3. RESIDENTIAL, PUBLIC AND INDUSTRIAL BUILDINGS AND STRUCTURES

#### GENERAL

**3.1** Buildings and structures shall be divided by aseismic joints in cases where:

- a building or structure has a complex configuration in plan view;

adjacent sections of a building or structure are different in height by 5 m and more; in one-storey houses up to 10 m, built in an area with a seismicity of 7, seismic joints shall be optional

**3.2** Aseismic joints shall divided buildings and structures over their entire height. The foundation may be without an aseismic joint, except when it coincides with the settlement joint.

**3.3** The spacing between aseismic joints and building height shall not exceed the values listed in Table 8.

**3.13** Balconies and their connections to floor panels shall be designed as cantilever beams or panels.

In buildings with masonry walls, the overhanging length of balconies shall not exceed 1.5 m.

**3.14** Foundations of buildings and structures in seismic areas shall be designed as per codes and rules (SNiP) covering foundation design.

**3.15** In seismic areas, a layer of grade 100 mortar at least 40 mm thick and longitudinal reinforcement with three, four and six 10 mm bars at a seismicity of 7, 8 and 9, respectively, shall be laid on top of prefabricated strip foundations. The longitudinal bars shall be interconnected with 6 mm transverse bars at 300 to 400 mm intervals.

## **SKELETON TYPE BUILDINGS**

**3.18** In skeleton-type buildings, the structural element subjected to a horizontal seismic load may be the frame itself, frame with its filling, frame with vertical braces, diaphragms or stiffeners.

**3.19** In skeleton-type buildings designed to a seismicity of 7 or 8, exterior masonry walls and interior reinforced concrete or steel frames (uprights) may be used, in which case the requirements established for stone buildings shall be met. The height of such buildings shall not exceed 7 m.

**3.20** The rigid joints of reinforced concrete frames shall be strengthened with the aid of welded wire fabric, spirals or closed binders.

Girder and column sections adjoining rigid joints of frames at a distance equal to one and a half of the height of their cross-section shall be strengthened using closed transverse reinforcement (binders) installed at calculated intervals but not less than 100 mm and, in framework with load-bearing diaphragms, 200 mm.

**3.21** Diaphragms, braces and stiffeners subjected to horizontal loads shall be continuous over the entire building height and extend uniformly and symmetrically from the building's center of mass.

**3.22** Light hanging panels may be used as enclosure walls in skeleton-type buildings. Brick or stone filling consistent with the requirements of paragraph 3.35 hereinbelow may be used.

**3.23** Nonbearing masonry walls shall be allowed when:

- frame uprights next to the wall are spaced not more than 6 m apart;
- the height of buildings on sites with a seismicity of 7, 8 and 9 does not exceed 18, 16 and 9 m, respectively.

**3.24** The masonry of nonbearing walls in skeleton-type buildings shall be of category I or II (as per paragraph 3.39 hereinbelow) and have flexible bracing with the frame, not hindering its horizontal movements along the walls.

A clearance of at least 20 mm shall be provided between the wall surface and frame uprights. Aseismic chords connected to the frame shall be provided along the entire wall length at the floor panel and window hole top levels.

At intersections of the end and transverse walls with longitudinal ones, aseismic joints shall be provided over the entire wall height.

**3.25** Stair wells and elevator shafts in skeleton-type buildings shall be conceived as single-storey built-in elements not affecting the rigidity of the frame or as a rigid core subjected to seismic loading.

In the case of skeleton-type buildings with up to five storeys, designed to a seismicity of 7 and 8, staircases and elevator shafts may be designed within the building's footprint yet separate from the framework. Designing staircases as stand-alone structures shall not be allowed.

**3.26** The load-bearing structural elements in high-rise buildings (with more than 16 storeys) shall be frames with diaphragms, bracing or stiffeners.

Given a choice of design concepts, preference shall be given to those in which the plasticity range is attained primarily in horizontal members of the frame (cross bars, tie bars, etc.).

**3.27** In addition to bending and shear strains in uprights, high-rise buildings shall also be designed taking into account axial strains, strain capacity of foundations and stability toward tilting forces.

**3.28** Erecting high-rise buildings and buildings as per Item 4 in Table 4 on sites with soils of category III (as per Table 1) shall not be allowed.

**3.29** The foundations of high-rise buildings on non-rocky soils shall, as a rule, be of the pile type or in the form of a solid foundation slab.

## **LARGE PANEL BUILDINGS**

**3.30** Large-panel buildings shall be designed with longitudinal and lateral walls connected with one another as well as with floor and roof panels to form a single seismically loaded three-dimensional system.

The following provisions shall be made in designing large-panel buildings:

- wall and floor panels shall, as a rule, be sized for single rooms;
- wall and floor panels shall be interconnected through welded protruding reinforcement bars, anchor rods and inserts with vertical shafts and joints being grouted with fine low slump concrete;
- when floor panels are laid on exterior walls and on walls next to expansion joints, reinforcement rods protruding from the former shall be welded to the vertical reinforcement of the latter.

**3.31** The reinforcement of wall panels shall be in the form of three-dimensional frames or wire mesh. In cases where sandwich panels are used as exterior walls, the thickness of the inner load-bearing concrete layer shall be at least 100 mm.

**3.32** Horizontal joints shall be capable of handling the design forces arising in them. The required cross-sectional area of steel bracing in joints between panels shall be determined by way of calculation but it shall not be less than 1 sq. cm per meter of joint length and, in the case of buildings with up to five storeys on sites with a seismicity of 7 and 8, not less than 0.5 sq. cm per meter of joint length. Not more than 65 % of the design vertical reinforcement may be placed at wall intersections.

**3.33** As a rule, walls shall be solid along the entire length and width of a building.

**3.34** Loggias shall, as a rule, be of the built-in type, equal in length to the spacing between adjacent walls. Reinforced concrete frames shall be arranged in the exterior wall plane to accommodate loggias.

No jut windows shall be allowed.